Pengzhi Gao

CONTACT INFORMATION

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Haidian District, Beijing, China 100085

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EDUCATION

Rensselaer Polytechnic Institute, Troy, NY, USA

Ph.D., Electrical Engineering, August 2013 - December 2017

University of Pennsylvania, Philadelphia, PA, USA

M.S., Electrical Engineering, September 2011 - May 2013

Xidian University, Xi'an, China

B.S. (with honors), Electronic and Information Engineering, August 2007 - July 2011

WORK EXPERIENCE

Senior Algorithm Engineer & Tech Lead Manager June 2024 to Present

Large Language Model Team,
Xiaomi, Inc., Beijing, China
Supervisor: Wei Liu and Jian Luan

- Led a cross-functional UI Agent & Agentic RL team of fifteen researchers, engineers, and interns to build native GUI Agent models that deliver one-step direct action capabilities in XiaoAi, enabling the assistant to autonomously complete multi-step GUI tasks from user instructions.
- Developed many-to-many multilingual machine translation models, GemmaX (NAACL 2025), which became the top trending model in Hugging Face's machine translation category, gaining over two hundred thousand downloads, and supporting real-time subtitle translation systems in XiaoAi Translate.

Staff Research & Development Engineer September 2020 to June 2024

Natural Language Processing Department,
Baidu, Inc., Beijing, China
Supervisor: Dr. Zhongjun He

- Developed a unified multilingual sentence representation model, MuSR, that supports more than 220 languages (EMNLP 2023 Industry Track), which is used as the foundation model for the similarity-based bitext mining system in Baidu Translate.
- Proposed a simple but effective training strategy to boost the zero-shot performance of multilingual neural machine translation (ACL 2023 Findings). Developed data processing and model training pipelines and improved the multilingual machine translation system that supports more than 200 languages in Baidu Translate.
- Designed Bi-SimCut, a simple but effective training strategy that boosts the performance of neural machine translation (NAACL 2022). Developed a training pipeline based on Bi-SimCut and improved the machine translation system in Baidu Translate.

Data Scientist / Machine Learning Engineer February 2018 to April 2020

Core Machine Learning Team,
Petuum, Inc., Pittsburgh, PA, USA
Supervisor: Dr. Tong Wen and Dr. Zhiting Hu

- Designed and implemented a machine learning library for Petuum AI Builder Platform.
- Designed and developed Texar-PyTorch (gaining over 745 stars on GitHub), an open-source machine learning and text generation toolkit based on PyTorch. Maintained and contributed to Texar (gaining over 2390 stars on GitHub), an open-source machine learning and text generation toolkit based on TensorFlow.
- Designed and developed Forte (gaining over 250 stars on GitHub), a toolkit for building natural language processing pipelines, featuring cross-task interaction, adaptable data-model interfaces and many more (EMNLP 2020 System Demonstrations).

Research Intern December 2010 to May 2011
Internet Media Group,
Microsoft Research Asia, Beijing, China
Supervisor: Dr. Feng Wu and Dr. Chong Luo
- Analyzed the data collected from 54 sensors deployed in Intel Berkeley Research Lab to exploit the temporal correlations in sensor readings. Developed a joint source network coding scheme for approximate data gathering in the wireless sensor network.

Skills

- **Research**: Generative AI, LLM/VLM-based Agents, Machine Learning, Natural Language Processing, Signal Processing, High-dimensional Statistics
- **Programming**: Python > C/C++ = Matlab > Java = R
- **Software**: PyTorch, TensorFlow, Keras, Scikit-learn, Linux, MacOS, Git

Honors and Awards

- North America Finalist of IBM Watson Build Challenge 2017
- Paper selected as runner-up for the Best Paper in Electric Energy Systems Track at Hawaii International Conference on System Sciences 2015
- Founders Award of Excellence (top 1%) 2015
- Paper selected as one of the Best Conference Papers on Power System Analysis and Modeling at IEEE Power & Energy Society General Meeting 2014
- Excellent Graduate of Xidian University (top 1%) 2011
- National Scholarship (top 0.2% nationwide) 2010
- First prize of the College Academic and Technological Scholarship (top 2%) 2008 - 2010
- Excellent Student Award (top 1%) 2008

Preprints & Technical Reports

* denotes equal contribution.

1. B. Han, Q. Wang, S. Liu, Z. Xiong, C. Wu, Y. Li, **P. Gao**, W. Liu, J. Luan, H. Qu, Y. Song, and Z. Cai. "Learning Reliable GUI Agents under Imperfect Priors." *arXiv: 2609.39547*, 2026.

2. T. Xiong, X. Hu, W. Wang, Q. Wu, C. Wu, **P. Gao**, W. Liu, J. Luan, and S. Zhang. "Task-Adaptive Rubrics for GUI Reward Modeling." *arXiv: 2608.24174*, 2026.

3. C. Han, **P. Gao**, P. Fu, and J. Luan. "Reference-Free Post-Training of Open Large Language Models for Multilingual Machine Translation." *arXiv: 2608.10812*, 2026.

4. W. Cao, C. Duan, P. Fu, **P. Gao**, N. Lian, F. Liu, H. Liu, H. Qu, Q. Wu, Z. Yu, T. Chen, S. Cui, A. Du, S. Jia, Y. Li, W. Liu, Y. Liu, W. Lu, Z. Luo, H. Sun, J. Sun, C. Tan, Y. Wang, C. Wu, T. Xiong, J. Yang, Y. Yuan, R. Zhang, S. Zhang, J. Zhu, J. Luan, and C. Zou. "Xiaomi-GUI-0 Technical Report." *arXiv: 2606.31410*, 2026.

5. G. Liu, J. Ye, **P. Gao**, W. Liu, J. Luan, Y. Liu, and Y. Li. "SimuWoB: Simulating Real-World Mobile Apps for Fast and Faithful GUI Agent Benchmarking." *arXiv: 2605.25160*, 2026.

6. W. Xu, K. Huang, Y. Feng, J. Li, Y. Chen, Y. Liu, Z. Jiang, H. Qu, **P. Gao**, W. Liu, J. Luan, X. Hu, and B. An. "How Mobile World Model Guides GUI Agents?" *arXiv: 2605.10347*, 2026.

7. Y. Chen*, **P. Gao***, Q. Wu, K. Huang, Y. Liu, H. Qu, K. Shao, and J. Luan. "Digital Agents Meet World Models: A Survey." 2026.

8. Y. Shang, **P. Gao**, Y. Yang, J. Ma, W. Liu, J. Luan, and J. Su. "ExPosST: Explicit Positioning with Adaptive Masking for LLM-Based Simultaneous Machine Translation." *arXiv: 2603.14903*, 2026.

9. Y. Shang*, **P. Gao***, W. Liu, J. Luan, and J. Su. "Scaling Model and Data for Multilingual Machine Translation with Open Large Language Models." *arXiv: 2602.11961*, 2026.

10. L. Gao, L. Zhang, S. Wang, **P. Gao**, W. Liu, J. Luan, S. Wang, Y. Li, M. Xu. "MobileViews: A Million-scale and Diverse Mobile GUI Dataset." *arXiv: 2409.14337*, 2024.

11. **P. Gao**, Z. He, H. Wu, and H. Wang. "Towards Boosting Many-to-Many Multilingual Machine Translation with Large Language Models." *arXiv: 2401.05861*, 2024.

12. R. Wang*, **P. Gao***, and M. Wang. "Robust Matrix Completion by Exploiting Dynamic Low-dimensional Structures." *DOI: 10.21203/rs.3.rs-420556/v1*, 2021.

13. Z. Hu, **P. Gao**, A. Bukkittu, and Z. Hu. "Introducing Texar-PyTorch: An ML Library integrating the best of TensorFlow into PyTorch." October, 2019.

Journal Publications

1. **P. Gao***, R. Wang*, M. Wang, and J. H. Chow. "Low-rank Matrix Recovery from Noisy, Quantized and Erroneous Measurements." *IEEE Transactions on Signal Processing*, 2018, 66 (11): 2918-2932.

2. **P. Gao**, M. Wang, J. H. Chow, M. Berger, and L. M. Seversky. "Missing Data Recovery for High-dimensional Signals with Nonlinear Low-dimensional Structures." *IEEE Transactions on Signal Processing*, 2017, 65 (20): 5421-5436.

3. **P. Gao**, M. Wang, J. H. Chow, S. G. Ghiocel, B. Fardanesh, G. Stefopoulos, and M. P. Razanousky. "Identification of Successive "Unobservable" Cyber Data Attacks in Power Systems Through Matrix Decomposition." *IEEE Transactions on Signal Processing*, 2016, 64 (21): 5557-5570.

4. **P. Gao**, M. Wang, S. G. Ghiocel, J. H. Chow, B. Fardanesh, and G. Stefopoulos. "Missing Data Recovery by Exploiting Low-dimensionality in Power System Synchrophasor Measurements." *IEEE Transactions on Power Systems*, 2016, 31 (2): 1006-1013.

Conference Publications

1. C. Zhang, W. Cao, L. Zhang, Y. Chen, **P. Gao**, W. Liu, J. Luan, C. Li, and L. Zou. "From Constraint to Guidance: Progressive Process Rewards for GUI Agents." *Proc. of the 20th ACM International Conference on Web Search and Data Mining (WSDM)*, 2027.

2. W. Xu, Z. Jiang, L. Jiang, K. Huang, Y. Liu, **P. Gao**, W. Liu, J. Luan, X. Hu, and B. An. "OfficeInstruct: A Dataset for Tool-Augmented Office Agents on Long-Trajectory Generative Tasks." *Proc. of the 40th Annual Conference on Neural Information Processing Systems (NeurIPS): Evaluations and Datasets Track*, 2026.

3. T. Xiong, X. Hu, Y. Chen, Y. Liu, C. Wu, **P. Gao**, W. Liu, J. Luan, and S. Zhang. "GUI-PRA: Process Reward Agents for GUI Tasks." *Proc. of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2026.

4. L. Zhang, Y. Chen, C. Zhang, W. Cao, K. Huang, **P. Gao**, W. Liu, J. Luan, C. Li, and L. Zou. "GSAR: Goal-State-Anchor Rewards for Mobile GUI Agents with Self-Evolving Data Synthesis." *Proc. of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2026.

5. H. Qu*, Y. Liu*, R. Jin, W. Zhang, **P. Gao**, W. Liu, and J. Luan. "Scaling, Benchmarking, and Reasoning of Vision-Language Agents for Mobile GUI Navigation." *Proc. of the 43rd International Conference on Machine Learning (ICML)*, 2026.

6. Y. Liu, W. Xu, K. Huang, C. Chen, J. Zhao, **P. Gao**, W. Liu, J. Luan, S. Shang, B. Du, J. Wen, and R. Yan. "CoME: Empowering Channel-of-Mobile-Experts with Informative Hybrid-Capabilities Reasoning." *Proc. of the 43rd International Conference on Machine Learning (ICML)*, 2026.

7. Q. Wu*, Z. Yang*, H. Li, **P. Gao**, W. Liu, and J. Luan. "MobileBench-OL: A Comprehensive Chinese Benchmark for Evaluating Mobile GUI Agents in Real-World Environment." *Findings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2026.

8. Y. Chen, Y. Liu, L. Zhang, **P. Gao**, J. Luan, and W. Liu. "STEP: Success-Rate-Aware Trajectory-Efficient Policy Optimization." *Findings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2026.

9. R. Jin, **P. Gao**, Y. Ren, Z. Han, T. Zhang, W. Huang, W. Liu, J. Luan, and D. Xiong. "Revisiting Entropy in Reinforcement Learning for Large Reasoning Models." *Findings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2026.

10. G. Liu, J. Ye, J. Liu, W. Liu, **P. Gao**, J. Luan, Y. Li, and Y. Liu. "Mobile GUI Agents under Real-world Threats: Are We There Yet?" *Proc. of the 24th ACM International Conference on Mobile Systems, Applications, and Services (MobiSys)*, 2026.

11. K. Huang, W. Xu, Y. Liu, Q. Wang, **P. Gao**, W. Liu, J. Luan, B. Wang, and B. An. "MobileIPL: Enhancing Mobile Agents Thinking Process via Iterative Preference Learning." *Proc. of the 14th International Conference on Learning Representations (ICLR)*, 2026.

12. W. Xu, Z. Jiang, Y. Liu, **P. Gao**, W. Liu, J. Luan, Y. Li, Y. Liu, B. Wang, and B. An. "SMAN-Bench: A Cross-System Benchmark for Mobile Agents under Single- and Multi-path, Ambiguous, and Noisy Tasks." *Proc. of the 14th International Conference on Learning Representations (ICLR)*, 2026.

13. L. Gao, L. Zhang, **P. Gao**, W. Liu, J. Luan, and M. Xu. "GUI-Shift: Enhancing VLM-Based GUI Agents through Self-supervised Reinforcement Learning." *Proc. of the 14th International Conference on Learning Representations (ICLR)*, 2026.

14. G. Liu, J. Ye, J. Liu, Y. Li, W. Liu, **P. Gao**, J. Luan, and Y. Liu. "Hijacking JARVIS: Benchmarking Mobile GUI Agents against Unprivileged Third Parties." *Proc. of the Workshop on Edge and Mobile Foundation Models (EdgeFM)*, 2025.

15. Q. Wu, **P. Gao**, W. Liu, and J. Luan. "BacktrackAgent: Enhancing GUI Agent with Error Detection and Backtracking Mechanism." *Proc. of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2025.

16. M. Cui*, **P. Gao***, W. Liu, J. Luan, and B. Wang. "Multilingual Machine Translation with Open Large Language Models at Practical Scale: An Empirical Study." *Proc. of the 2025 Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL)*, 2025.

17. M. Sun, W. Liu, J. Luan, **P. Gao**, and B. Wang. "Mixture of Diverse Size Experts." *Proc. of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP): Industry Track*, 2024.

18. J. Du, Y. Wang, W. Zhao, Z. Deng, S. Liu, R. Lou, H. P. Zou, P. N. Venkit, N. Zhang, M. Srinath, H. R. Zhang, V. Gupta, Y. Li, T. Li, F. Wang, Q. Liu, T. Liu, **P. Gao**, C. Xia, C. Xing, J. Cheng, Z. Wang, Y. Su, R. S. Shah, R. Guo, J. Gu, H. Li, K. Wei, Z. Wang, L. Cheng, S. Ranathunga, M. Fang, J. Fu, F. Liu, R. Huang, E. Blanco, Y. Cao, R. Zhang, P. S. Yu, and W. Yin. "LLMs assist NLP Researchers: Critique Paper (Meta-)Reviewing." *Proc. of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2024.

19. **P. Gao**, R. Zhang, Z. He, H. Wu, and H. Wang. "An Empirical Study of Consistency Regularization for End-to-End Speech-to-Text Translation." *Proc. of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 2024.

20. **P. Gao**, L. Zhang, Z. He, H. Wu, and H. Wang. "Learning Multilingual Sentence Representations with Cross-lingual Consistency Regularization." *Proc. of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP): Industry Track*, 2023.

21. **P. Gao**, L. Zhang, Z. He, H. Wu, and H. Wang. "Improving Zero-shot Multilingual Neural Machine Translation by Leveraging Cross-lingual Consistency Regularization." *Findings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*, 2023.

22. **P. Gao**, Z. He, H. Wu, and H. Wang. "Bi-SimCut: A Simple Strategy for Boosting Neural Machine Translation." *Proc. of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 2022.

23. J. Li, **P. Gao**, X. Wu, Y. Feng, Z. He, H. Wu, and H. Wang. "Mixup Decoding for Diverse Machine Translation." *Findings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2021.

24. R. Wang, T. Chen, Z. Xu, and **P. Gao**. "Robust Low-Rank Tensor Recovery From Quantized and Corrupted Measurements." *Proc. of Asilomar Conference on Signals, Systems, and Computers*, 2021.

25. Z. Liu, G. Ding, A. Bukkittu, M. Gupta, **P. Gao**, A. Ahmed, S. Zhang, X. Gao, S. Singhavi, L. Li, W. Wei, Z. Hu, H. Shi, X. Liang, T. Mitamura, E. P. Xing, and Z. Hu. "A Data-Centric Framework for Composable NLP Workflows." *Proc. of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP): System Demonstrations*, 2020.

26. M. Wang, J. H. Chow, Y. Hao, S. Zhang, W. Li, R. Wang, **P. Gao**, C. Lackner, E. Farantatos, and M. Patel. "A Low-rank Framework of PMU Data Recovery and Event Identification." *Proc. of the First IEEE International Conference on Smart Grid Synchronized Measurements and Analytics (SGSMA)*, 2019.

27. G. M. De Mijolla, S. Konstantinopoulos, **P. Gao**, J. H. Chow, and M. Wang. "An Evaluation of Algorithms for Synchrophasor Missing Data Recovery." *Proc. of the 20th Power Systems Computation Conference (PSCC)*, 2018.

28. **P. Gao** and M. Wang. "Dynamic Matrix Recovery from Partially Observed and Erroneous Measurements." *Proc. of the International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2018.

29. M. Wang, J. H. Chow, **P. Gao**, Y. Hao, W. Li, and R. Wang. "Recent Results of PMU Data Analytics by Exploiting Low-dimensional Structures." *Proc. of the 10th Bulk Power Systems Dynamics and Control Symposium (IREP)*, 2017.

30. **P. Gao**, R. Wang, and M. Wang. "Quantized Low-rank Matrix Recovery with Erroneous Measurements: Application to Data Privacy in Power Grids." *Proc. of Asilomar Conference on Signals, Systems, and Computers*, 2016.

31. **P. Gao**, M. Wang, and J. H. Chow. "Matrix Completion with Columns in Union and Sums of Subspaces." *Proc. of IEEE Global Conference on Signal and Information Processing (GlobalSIP)*, 2015.

32. M. Wang, J. H. Chow, **P. Gao**, X. T. Jiang, Y. Xia, S. G. Ghiocel, B. Fardanesh, G. Stefopoulos, Y. Kokai, N. Saito, and M. P. Razanousky. "A Low-Rank Matrix Approach for the Analysis of Large Amounts of Power System Synchrophasor Data." *Proc. of Hawaii International Conference on System Sciences (Runner-up of Best Paper in Electric Energy Systems Track)*, 2015.

33. M. Wang, **P. Gao**, S. G. Ghiocel, J. H. Chow, B. Fardanesh, G. Stefopoulos, and M. P. Razanousky. "Identification of "Unobservable" Cyber Data Attacks on Power Grids." *Proc. of IEEE SmartGridComm*, 2014.

34. **P. Gao**, M. Wang, S. G. Ghiocel, and J. H. Chow. "Modeless Reconstruction of Missing Synchrophasor Measurements." *Proc. of IEEE Power & Energy Society General Meeting (Best Conference Papers on Power System Analysis and Modeling)*, 2014.

Projects

Forte: A Data-Centric Framework for Composable NLP Workflows
Core Machine Learning Team,
Petuum, Inc., Pittsburgh, PA, USA

- Forte is a toolkit for building NLP pipelines, featuring composable components, convenient data interfaces, and cross-task interaction. Forte designs a universal data representation format for text, making it a one-stop platform to assemble state-of-the-art NLP/ML technologies, ranging from Information Retrieval and Natural Language Understanding to Natural Language Generation. Forte was originally developed at CMU and is actively contributed by Petuum in collaboration with other institutes. I am one of the main contributors to Forte.

Texar-PyTorch: A Modularized, Versatile, and Extensible Toolkit for Text Generation
Core Machine Learning Team,
Petuum, Inc., Pittsburgh, PA, USA

- Texar-PyTorch is a toolkit aiming to support a broad set of machine learning, especially natural language processing and text generation tasks. Texar-PyTorch provides a library of easy-to-use ML modules and functionalities for composing whatever models and algorithms. The tool is designed for both researchers and practitioners for fast prototyping and experimentation. Texar-PyTorch was originally developed and is actively contributed by Petuum and CMU in collaboration with other institutes. I am one of the main contributors to Texar-PyTorch.

Professional Activities & Service

- Member of IEEE, 2018 - present. Member of ACL, 2022 - present.
- Area Chair:
  The Association for Computational Linguistics Rolling Review (ACL ARR).
- Journal Reviewer:
  IEEE Transactions on Signal Processing,
  IEEE Transactions on Audio, Speech and Language Processing,
  IEEE Transactions on Smart Grid,
  IEEE Transactions on Automatic Control,
  IEEE Transactions on Power Delivery,
  IEEE/ACM Transactions on Networking,
  IEEE Signal Processing Letters,
  Annals of Mathematics and Artificial Intelligence,
  Journal of Data-centric Machine Learning Research (DMLR).
- Program Committee Member / Conference Reviewer:
  The Conference on Uncertainty in Artificial Intelligence (UAI),
  The Annual AAAI Conference on Artificial Intelligence (AAAI),
  The Conference on Neural Information Processing Systems (NeurIPS),
  The International Conference on Learning Representations (ICLR),
  The International Conference on Machine Learning (ICML),
  The Association for Computational Linguistics Rolling Review (ACL ARR),
  The Annual Meeting of the Association for Computational Linguistics (ACL),
  The Conference on Empirical Methods in Natural Language Processing (EMNLP),
  The Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL),
  The Conference of the European Chapter of the Association for Computational Linguistics (EACL),
  The International Conference on Computational Linguistics (COLING),
  The Conference on Language Modeling (COLM),
  The China National Conference on Computational Linguistics (CCL),
  The CCF International Conference on Natural Language Processing and Chinese Computing (NLPCC),
  SIAM International Conference on Data Mining (SDM),
  The ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD),
  IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR),
  IEEE/CVF International Conference on Computer Vision (ICCV),
  The European Conference on Computer Vision (ECCV),
  British Machine Vision Conference (BMVC),
  IEEE International Conference on Communications, Control, and Computing Technologies

for Smart Grids (SmartGridComm),
American Control Conference (ACC),
Intelligent System Applications to Power Systems Conference (ISAP),
International Conference on Intelligent Multilingual Information Processing (IMLIP).